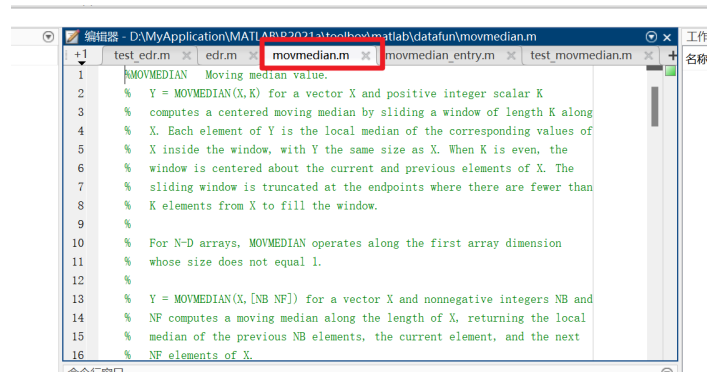
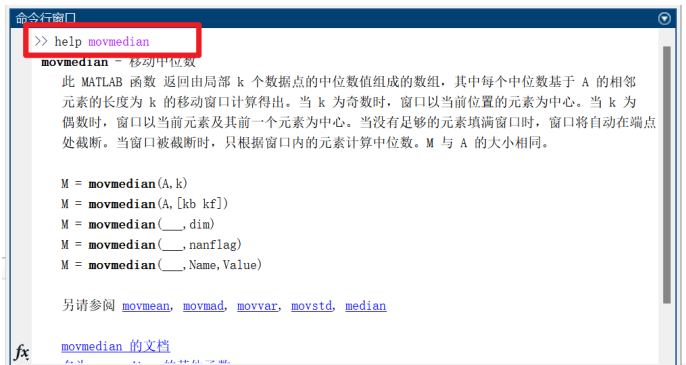


Matlab 转 c代码 为python所用 SOP



本文档以函数movmedian为例，详细记录转出流程

1. 检查函数是否存在，若不存在则先下载



- 若不存在，则会提示下载链接，点击下载

2. 套壳（自定义函数中调用目标函数）

```
1 function output = movmedian_entry(X, K) %#codegen
2     % Validate inputs
3     assert(isnumeric(X) && isvector(X), 'X must be a numeric vector');
4     assert(isnumeric(K) && (isscalar(K) || numel(K) == 2), ...
5         'K must be a scalar or a two-element vector');
6
7     % 计算移动中值
8     output = movmedian(X, K);
9 end
10
```

3. 使用coder插件生成.c文件

```
1 codegen movmedian_entry -args {coder.typeof(0, [1, Inf]), coder.typeof(0, [1, 2])} -config:dll -report
```

- 其中，`coder.typeof` 表示参数的数据格式，`[1, Inf]`代表 1行*n列，`[1, 2]`代表 1行*2列；
- 若`row`和`col`都不是确定的，则使用`coder.typeof(0, [inf, Inf], [1, 1])`
- 生成的文件结构树：

```
1 D:\matlab\codegen\dll\movmedian_entry>tree /f
2 卷 Data 的文件夹 PATH 列表
3 卷序列号为 FC70-DD4A
4 D:.
5 | .gitignore
6 | buildInfo.mat
7 | codedescriptor.dmr
8 | codeInfo.mat
9 | compileInfo.mat
10 | defines.txt
11 | lccstub.obj
12 | movmedian_entry.c
13 | movmedian_entry.def
14 | movmedian_entry.dll
15 | movmedian_entry.exp
16 | movmedian_entry.h
17 | movmedian_entry.lib
18 | movmedian_entry.obj
19 | movmedian_entry_data.c
20 | movmedian_entry_data.h
21 | movmedian_entry_data.obj
22 | movmedian_entry_emxAPI.c
23 | movmedian_entry_emxAPI.h
24 | movmedian_entry_emxAPI.obj
25 | movmedian_entry_emxutil.c
26 | movmedian_entry_emxutil.h
27 | movmedian_entry_emxutil.obj
28 | movmedian_entry_initialize.c
29 | movmedian_entry_initialize.h
30 | movmedian_entry_initialize.obj
31 | movmedian_entry_rtw.bat
32 | movmedian_entry_rtw.mk
33 | movmedian_entry_rtw.rsp
34 | movmedian_entry_rtw_ref.rsp
35 | movmedian_entry_terminate.c
36 | movmedian_entry_terminate.h
37 | movmedian_entry_terminate.obj
38 | movmedian_entry_types.h
39 | rtGetInf.c
40 | rtGetInf.h
```

```

41 | rtGetInf.obj
42 | rtGetNaN.c
43 | rtGetNaN.h
44 | rtGetNaN.obj
45 | rtwtypes.h
46 | rtw_proj.tmw
47 | rt_nonfinite.c
48 | rt_nonfinite.h
49 | rt_nonfinite.obj
50 | SortedBuffer.c
51 | SortedBuffer.h
52 | SortedBuffer.obj
53 | _clang-format
54 |
55 | └─examples
56 |     main.c
57 |     main.h
58 |
59 | └─html
60 |     report.mldatx
61 |
62 | └─interface
63 |     _coder_movmedian_entry_api.c
64 |     _coder_movmedian_entry_api.h
65 |     _coder_movmedian_entry_info.c
66 |     _coder_movmedian_entry_info.h
67 |     _coder_movmedian_entry_mex.c
68 |     _coder_movmedian_entry_mex.h

```

4. 新建c项目

- 将上一步生成的所有.c 和 .h文件(除了interface文件夹下的文件) 复制到空项目的src文件夹下
- 将matlab下载文件夹里的tmwtypes.h (".....\MATLAB\R2021a\extern\include\tmwtypes.h") 也复制到src中（否则main.h会报错，找不到该文件）
- 编写CMakeLists.txt:

```

1 cmake_minimum_required (VERSION 3.2)
2
3 set(CMAKE_C_COMPILER "D:\\MyApplication\\msys64\\ucrt64\\bin\\gcc.exe")
4 set(CMAKE_CXX_COMPILER "D:\\MyApplication\\msys64\\ucrt64\\bin\\g++.exe")
5
6 project(movmedian)
7
8 aux_source_directory(./src/ DIR_SRCS)

```

```
9
10 add_library(movmedian SHARED ${DIR_SRCS})
11 target_link_libraries(movmedian m)
12
```

- 项目树如下：

```
1 C:\Users\Hzl\Desktop\matlab\matlab-computing-algorithm\movmedian>tree /f
2 卷 Windows 的文件夹 PATH 列表
3 卷序列号为 FA2A-3B4A
4 C:
5 |   CMakeLists.txt
6 |   test_movmedian.py
7 |   └─src
8       main.c
9       main.h
10      movmedian_entry.c
11      movmedian_entry.h
12      movmedian_entry_data.c
13      movmedian_entry_data.h
14      movmedian_entry_emxAPI.c
15      movmedian_entry_emxAPI.h
16      movmedian_entry_emxutil.c
17      movmedian_entry_emxutil.h
18      movmedian_entry_initialize.c
19      movmedian_entry_initialize.h
20      movmedian_entry_terminate.c
21      movmedian_entry_terminate.h
22      movmedian_entry_types.h
23      rtGetInf.c
24      rtGetInf.h
25      rtGetNaN.c
26      rtGetNaN.h
27      rtwtypes.h
28      rtw_proj.tmw
29      rt_nonfinite.c
30      rt_nonfinite.h
31      SortedBuffer.c
32      SortedBuffer.h
```

5. 修改main.h

- 原来的代码：

```

1 #ifndef MAIN_H
2 #define MAIN_H
3
4 /* Include Files */
5 #include "rtwtypes.h"
6 #include <stddef.h>
7 #include <stdlib.h>
8
9 #ifdef __cplusplus
10 extern "C" {
11 #endif
12
13 /* Function Declarations */
14 extern int main(int argc, char **argv);
15
16 #ifdef __cplusplus
17 }
18 #endif
19
20 #endif

```

- 修改后的代码：

```

1 #ifndef MAIN_H
2 #define MAIN_H
3
4 /* Include Files */
5 #include "rtwtypes.h"
6 #include <stddef.h>
7 #include <stdlib.h>
8 #include <stdio.h>
9
10 #ifdef __cplusplus
11 extern "C" {
12 #endif
13
14 /* Function Declarations */
15 void movmedian_wrapper( double* input_array, int rows, int cols,
16                        double* k_values, int k_size,
17                        double* output_array);
18
19 #ifdef __cplusplus
20 }
21 #endif
22

```

- main主函数没了，改成movmedian_wrapper（随意取名，之后python调用的时候使用该方法名）
- 其中input_array、k_values是函数的入参，rows，cols是input_array的大小，k_size是k_values的大小
- 将返回的数放在**double*** output_array中，以指针的形式返回，故方法的返回类型是void

6. 修改main.c

- 直接上修改后的代码：

```
1 #include "main.h"
2 #include "movmedian_entry.h"
3 #include "movmedian_entry_emxAPI.h"
4 #include "movmedian_entry_terminate.h"
5 #include "rt_nonfinite.h"
6 /* Function Declarations */
7 static double argInit_real_T(double);
8
9 static emxArray_real_T *c_argInit_UnboundedUnbounded_r(double *, int, int);
10
11 /* Function Definitions */
12 /*
13  * Arguments      : double
14  * Return Type    : double
15  */
16 static double argInit_real_T(double value)
17 {
18     return value;
19 }
20
21 /*
22  * Arguments      : double *, int, int
23  * Return Type    : emxArray_real_T *
24  */
25 static emxArray_real_T *c_argInit_UnboundedUnbounded_r(double *x, int rows,
    int cols)
26 {
27     emxArray_real_T *result;
28     double *result_data;
29     int idx0;
30     int idx1;
31     /* Set the size of the array.
32     Change this size to the value that the application requires. */
```

```

33     result = emxCreate_real_T(rows, cols);
34     result_data = result->data;
35     /* Loop over the array to initialize each element. */
36     for (idx0 = 0; idx0 < result->size[0U]; idx0++) {
37         for (idx1 = 0; idx1 < result->size[1U]; idx1++) {
38             /* Set the value of the array element.
39 Change this value to the value that the application requires. */
40             result_data[idx0 + result->size[0] * idx1] = argInit_real_T(x[idx0 +
41 result->size[0] * idx1]);
42         }
43     }
44     return result;
45
46 /* Wrapper function to expose to Python */
47 void movmedian_wrapper(double* input_array, int rows, int cols,
48                        double* k_values, int k_size,
49                        double* output_array) {
50
51     // emxArray_real_T* X = emxCreateWrapper_real_T((double*)input_array,
52 rows, cols);
53     emxArray_real_T* X = c_argInit_UnboundedUnbounded_r(input_array, rows,
54 cols);
55
56     emxArray_real_T* output = emxCreate_real_T(rows, cols);
57
58     /* Call the entry-point function */
59     movmedian_entry(X, (double*)k_values, output);
60
61     /* Copy result to output_array */
62     for (int i = 0; i < rows * cols; i++) {
63         output_array[i] = output->data[i];
64     }
65
66     /* Clean up */
67     emxDestroyArray_real_T(output);
68     emxDestroyArray_real_T(X);
69 }

```

- **c_argInit_UnboundedUnbounded_r** 函数

- 作用：copy一份输入数据到c的内存空间中，方便后续函数使用。
- 返回emxArray_real_T：是matlab coder生成的固定输入数据结构

- **movmedian_wrapper** 函数

- a. 处理输入的参数，变成emxArray_real_T 格式

- b. 调用目标方法 `movmedian_entry`
- c. 组装结果到 `output_array` 地址中
- `argInit_real_T` 函数
 - 这里没啥用...

7. 编写python测试文件

- 在项目下新建 `test_movmedian.py`:

```
1 import ctypes
2 import numpy as np
3
4 def movmedian_wrapper(input_array, k_values):
5     # 加载 DLL (确保 libmovmedian.dll 在当前目录或系统 PATH 中)
6     movmedian = ctypes.CDLL('./libmovmedian.dll')
7
8     # 定义函数的参数类型和返回类型
9     movmedian.movmedian_wrapper.argtypes = [
10         np.ctypeslib.ndpointer(dtype=np.float64, ndim=2, flags='C'), #
11         input_array
12         ctypes.c_int, # rows
13         ctypes.c_int, # cols
14         np.ctypeslib.ndpointer(dtype=np.float64, ndim=1, flags='C'), #
15         k_values
16         ctypes.c_int, # k_size
17         np.ctypeslib.ndpointer(dtype=np.float64, ndim=2, flags='C') #
18         output_array
19     ]
20     movmedian.movmedian_wrapper.restype = None
21
22     # 准备输出数组
23     output_array = np.zeros_like(input_array, dtype=np.float64)
24
25     # 调用 movmedian_wrapper 函数
26     movmedian.movmedian_wrapper(
27         input_array,
28         input_array.shape[0],
29         input_array.shape[1],
30         k_values,
31         k_values.size,
32         output_array
33     )
34
35     return output_array
```



```

34 if __name__ == '__main__':
35     # 创建示例输入数据
36     data = np.array([[4, 9, 6, -1, -2, -3, -1, 3, 4, 5]], dtype=np.float64)
37     print("Input Data:", data.tolist())
38
39     # 定义滑动窗口大小
40     k_values = np.array([2,0], dtype=np.float64) # 窗口大小为 2
41
42     # 调用 movmedian_wrapper 并打印输出
43     result = movmedian_wrapper(data, k_values)
44     print("Output Data:", result.tolist())
45

```

8. 编译与生成动态库

- 在项目根目录执行cmake:

```

1 PS C:\Users\Hzl\Desktop\matlab\matlab-computing-algorithm\movmedian> cmake .
2 CMake Deprecation Warning at CMakeLists.txt:1 (cmake_minimum_required):
3   Compatibility with CMake < 3.10 will be removed from a future version of
4   CMake.
5
6   Update the VERSION argument <min> value or use a ...<max> suffix to tell
7   CMake that the project does not need compatibility with older versions.
8
9
10 -- Configuring done (0.4s)
11 -- Generating done (0.1s)
12 -- Build files have been written to: C:/Users/Hzl/Desktop/matlab/matlab-
    computing-algorithm/movmedian

```

- 执行make:

```

1 PS C:\Users\Hzl\Desktop\matlab\matlab-computing-algorithm\movmedian> mingw32-
    make
2 [ 8%] Building C object CMakeFiles/movmedian.dir/src/main.c.obj
3 [ 16%] Linking C shared library libmovmedian.dll
4 [100%] Built target movmedian

```

- 最后生成一坨文件，但是都不重要，最重要的是dll文件😁

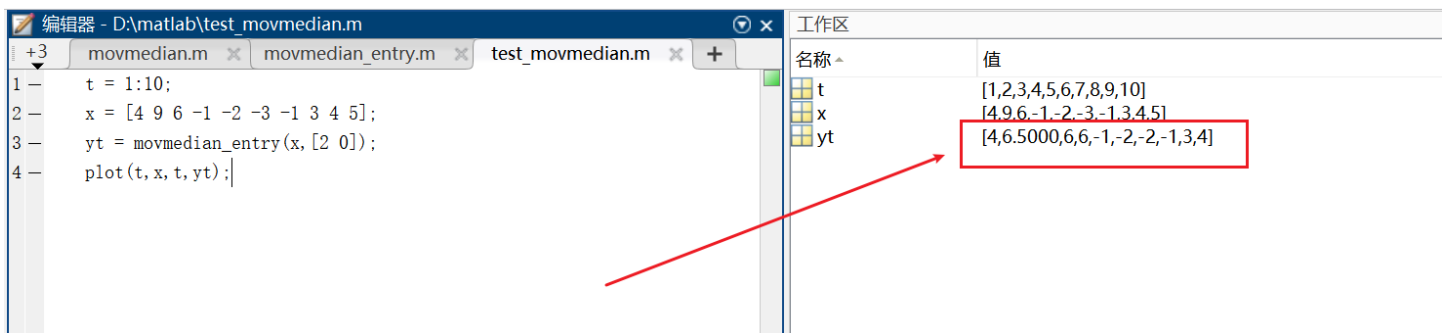
libmovmedian.dll
libmovmedian.dll.a

9. 测试

- 执行python脚本测试结果与matlab对比

```
1 C:\Users\Hzl\Desktop\matlab\matlab-computing-algorithm\movmedian>python  
  test_movmedian.py  
2 Input Data: [[4.0, 9.0, 6.0, -1.0, -2.0, -3.0, -1.0, 3.0, 4.0, 5.0]]  
3 Output Data: [[4.0, 6.5, 6.0, 6.0, -1.0, -2.0, -2.0, -1.0, 3.0, 4.0]]
```

- Matlab 结果:



🎉 完全一致，完结撒花